

**Supplemental Materials for A Dual-Channel Competition Framework Reveals Hidden Hydraulic
State Transitions**

Huayang Cai, Ph.D.,¹ Gaojin Li,² Ping Zhang, Ph.D.,³ Jianliang Lin, Ph.D.,⁴ Tongtiegang Zhao, Ph.D.,⁵
and Kairong Lin, Ph.D.⁶

¹ Professor, Institute of Estuarine and Coastal Research, School of Ocean Engineering and Technology,
Sun Yat-sen University / Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai),
Zhuhai, Guangdong 519082, China; State and Local Joint Engineering Laboratory of Estuarine Hydraulic
Technology / Guangdong Provincial Engineering Research Center of Coasts, Islands and Reefs /
Guangdong Provincial Key Laboratory of Marine Resources and Coastal Engineering; email:
caihy7@mail.sysu.edu.cn (Corresponding author)

² Ph.D. Student, Institute of Estuarine and Coastal Research, School of Ocean Engineering and
Technology, Sun Yat-sen University, Zhuhai, Guangdong 519082, China.

³ Research Fellow, State Key Laboratory of Estuarine and Coastal Research, East China Normal
University, Shanghai 200241, China.

⁴ Research Fellow, Institute of Estuarine and Coastal Research, School of Ocean Engineering and
Technology, Sun Yat-sen University, Zhuhai, Guangdong 519082, China.

⁵ Professor, School of Civil Engineering, Sun Yat-sen University, Zhuhai, Guangdong 519082, China.

⁶ Professor, School of Civil Engineering, Sun Yat-sen University, Zhuhai, Guangdong 519082, China.

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24 The Supplemental Materials provide the analytical derivations and numerical verification supporting the
25 hydraulic decomposition presented in the main manuscript. Text S1 develops the exact velocity-field
26 formulation in bipolar coordinates. Text S2 derives the corresponding scalar discharge expressions and
27 establishes the integrated confinement and release channels. Text S3 describes the numerical assessment
28 of local and integrated channel nonnegativity. Text S4 documents the numerical treatment of the spectral
29 integrals, algebraic consistency checks, convergence assessment, refinement of the characteristic filling
30 states, selection of the equal-discharge states, and evaluation of the corresponding wall-shear
31 distributions.

32 **Text S1. Exact Driving–Confinement–Release Formulation**

33 Consider a circular pipe of radius R_p containing a Newtonian liquid of dimensional depth h . The
34 dimensionless filling ratio H and filling angle α are related by

$$35 \quad H = h/R_p = 1 - \cos\alpha, \quad 0 < \alpha < \pi. \quad (S1)$$

36 Cartesian coordinates x and y are measured from the midpoint of the free-surface chord, with y positive
37 upward. The dimensionless coordinates are $X = x/R_p$ and $Y = y/R_p$; hence, the free surface lies at $Y =$
38 0 and the pipe center at $Y = \cos\alpha$. The dimensional streamwise velocity u is normalized according to

$$39 \quad U = \frac{\mu u}{\rho_f g R_p^2 S_h}, \quad (S2)$$

40 where μ is the dynamic viscosity, ρ_f is the fluid density, g is the gravitational acceleration, and S_h is the
41 imposed streamwise hydraulic gradient. The dimensionless physical velocity U_{FS} satisfies

$$-\nabla_{\perp}^2 U_{\text{FS}} = 1 \text{ in } \Omega, \quad U_{\text{FS}} = 0 \text{ on } \Gamma_w, \quad \frac{\partial U_{\text{FS}}}{\partial n} = 0 \text{ on } \Gamma_s, \quad (\text{S3})$$

where $\Omega(H)$ denotes the wetted cross section, Γ_w is the wetted circular wall, Γ_s is the horizontal shear-free surface, and n is the outward unit normal.

For analytical evaluation, the circular-segment domain is mapped to bipolar coordinates ξ and s , with $-\infty < \xi < \infty$ and $0 \leq s \leq 1$, through

$$X = \frac{\sin \alpha \sinh \xi}{\cosh \xi + \cos(\alpha s)}, \quad Y = -\frac{\sin \alpha \sin(\alpha s)}{\cosh \xi + \cos(\alpha s)}. \quad (\text{S4})$$

Under this transformation, $s = 0$ represents the horizontal free-surface chord and $s = 1$ represents the wetted circular wall. The coordinate ξ extends along the transformed domain, and the symmetry of the physical cross section corresponds to even dependence on ξ . For consistency with the pipe-centered geometry shown in Fig. 2 of the main manuscript, the vertical plotting coordinate is obtained as $Y - \cos \alpha$. This translation leaves the velocity fields and cross-sectional integrals unchanged.

The decomposition begins with the Poisson driving field U_D ,

$$U_D = \frac{1}{4} [1 - X^2 - (Y - \cos \alpha)^2] = \frac{\sin \alpha \sin [\alpha(1 - s)]}{2[\cosh \xi + \cos(\alpha s)]}. \quad (\text{S5})$$

The harmonic confinement field U_C supplies the correction required to impose a stationary condition on the auxiliary horizontal chord:

$$U_C = -\sin^2 \alpha \int_0^\infty \frac{k \sinh [k\alpha(1 - s)] \cos(k\xi)}{\sinh(\pi k) \sinh(\alpha k)} dk. \quad (\text{S6})$$

The mixed-boundary correction associated with the shear-free condition is

$$\chi_N = -\frac{\sin(2\alpha)}{2} \int_0^\infty \frac{\sinh [k\alpha(1 - s)] \cos(k\xi)}{\sinh(\pi k) \cosh(\alpha k)} dk. \quad (\text{S7})$$

60 The release field U_R is defined as the velocity recovered when the stationary auxiliary chord is replaced
 61 by the shear-free surface:

$$62 \quad U_R = \chi_N - U_C. \quad (S8)$$

63 The physical velocity therefore satisfies the exact field closure

$$64 \quad U_{FS} = U_D + U_C + U_R. \quad (S9)$$

65 Eqs. (S5)–(S9) distinguish three contributions. The field U_D provides the filling-dependent Poisson
 66 baseline, U_C imposes rigid-chord confinement, and U_R represents the subsequent release produced by
 67 replacing the stationary chord with the shear-free boundary.

68 At half filling, $\alpha = \pi/2$ and $\sin(2\alpha) = 0$. Consequently, χ_N vanishes and

$$69 \quad U_R = -U_C, \quad U_{FS} = U_D. \quad (S10)$$

70 Thus, equality between the physical and Poisson velocity fields at half filling results from exact
 71 cancellation between two finite boundary contributions rather than from the disappearance of boundary
 72 activity.

73 **Text S2. Exact Scalar Discharge Expressions and Boundary-Channel Construction**

74 For a velocity component U_j , define the associated dimensionless discharge contribution as

$$75 \quad Q_j = \iint_{\Omega} U_j \, dA^*, \quad dA^* = \frac{dA}{R_p^2}. \quad (S11)$$

76 Integration of the exact field closure in Eq. (S9) gives

$$77 \quad Q_{FS} = Q_D + Q_C + Q_R. \quad (S12)$$

78 The Poisson-baseline discharge is

$$Q_D(\alpha) = \frac{\alpha}{8} - \frac{\sin(2\alpha)}{12} + \frac{\sin(4\alpha)}{96}. \quad (\text{S13})$$

To express the discharge associated with the auxiliary rigid-chord problem, define

$$I_D(\alpha) = \int_0^\infty \frac{k^3}{\tanh(\alpha k) \sinh^2(\pi k)} dk, \quad (\text{S14})$$

The corresponding circular-segment discharge function is

$$Q_{\text{seg}}(\alpha) = \frac{\cot \alpha}{6} + \frac{\alpha}{8 \sin^4 \alpha} - \frac{\cos \alpha (5 - 2 \cos^2 \alpha)}{24 \sin^3 \alpha} - \pi I_D(\alpha). \quad (\text{S15})$$

After restoring the filling-dependent geometric scale, the rigid-chord discharge becomes

$$Q_{\text{RL}}(\alpha) = \sin^4 \alpha Q_{\text{seg}}(\alpha), \quad (\text{S16})$$

The confinement contribution is consequently

$$Q_C = Q_{\text{RL}} - Q_D. \quad (\text{S17})$$

For the physical shear-free problem, define

$$I_N(\alpha) = \int_0^\infty \frac{k \tanh(\alpha k)}{\sinh^2(\pi k)} dk. \quad (\text{S18})$$

The exact physical discharge is

$$Q_{\text{FS}}(\alpha) = \frac{\alpha - \sin(2\alpha) + \frac{1}{4} \sin(4\alpha) + 2\pi \sin^2(2\alpha) I_N(\alpha)}{8}, \quad (\text{S19})$$

The release contribution follows from the difference between the physical and rigid-chord states:

$$Q_R = Q_{\text{FS}} - Q_{\text{RL}}. \quad (\text{S20})$$

Because Q_C represents a suppressive correction, the nonnegative integrated confinement magnitude is

defined as $C_Q = -Q_C$. The integrated release magnitude is $R_Q = Q_R$. Hence,

$$C_Q = -Q_C = Q_D - Q_{RL}, R_Q = Q_R = Q_{FS} - Q_{RL}. \quad (S21)$$

Their signed difference is

$$S_B = R_Q - C_Q = Q_{FS} - Q_D, \quad (S22)$$

Eq. (S22) establishes that S_B is the net boundary-induced modification of the filling-dependent Poisson discharge. It is therefore distinct from the total physical discharge $Q_F S$. Substitution of C_Q and R_Q into the state and performance definitions in the main manuscript yields the integrated boundary activity, signed dominance, interaction coordinate, competition factor, and boundary cancellation-adjusted diagnostic.

At half filling, $\alpha = \pi/2$, and Eqs. (S13) and (S19) give $Q_D = Q_F S = \pi/16$. Eq. (S22) then gives $S_B = 0$, while Eq. (S21) gives $C_Q = R_Q$. The half-filled state is therefore an exact integrated balance between finite confinement and release contributions.

Text S3. Numerical Verification of Local Channel Nonnegativity

The two-channel representation requires each declared channel magnitude to remain nonnegative over the investigated hydraulic states. The local quantities examined in the partially filled conduit were

$$C = -U_C, \quad R = U_R, \quad J_+^{(U)} = U_D + U_R \quad (S23)$$

The assessment was performed for 100 uniformly spaced filling ratios over $0.01 \leq H \leq 1.99$. Because the exact fields are even in ξ , only the half-domain $0 \leq \xi \leq 8$ was evaluated. The computational grid contained 241 points in ξ and 101 points in s over $0 \leq s \leq 1$.

The spectral integrals in Eqs. (S6) and (S7) were evaluated over $10^{-6} \leq k \leq 12$ using 6,001 uniformly spaced points and trapezoidal integration. At each filling ratio, the minimum sampled values of C , R , and $J_+^{(U)}$ were recorded over the complete computational grid.

The Jacobian of the bipolar-coordinate transformation is

$$J(\xi, s) = \frac{\alpha \sin^2 \alpha}{[\cosh \xi + \cos(\alpha s)]^2}, \quad (S24)$$

and the dimensionless area element is

$$dA^* = J(\xi, s) d\xi ds. \quad (S25)$$

Accounting for symmetry about $\xi = 0$, the corresponding integrated channel magnitudes were evaluated as

$$C_Q = \iint_{\Omega} C dA^*, \quad R_Q = \iint_{\Omega} R dA^*, \quad J_{+,Q}^{(U)} = \iint_{\Omega} J_+^{(U)} dA^*. \quad (S26)$$

No negative values beyond numerical roundoff were detected for C , R , or $J_+^{(U)}$ over the sampled filling ratios and computational domain. Their cross-sectional integrals also remained nonnegative throughout the investigated range. In addition, C_Q and R_Q became equal at $H = 1$, in agreement with the exact half-filled relation obtained from Eq. (S22).

The calculation establishes numerical admissibility of the selected channels over the stated filling range and spatial discretization. It does not constitute an independent analytical proof of pointwise nonnegativity outside the investigated domain.

Text S4. Numerical Evaluation, Equal-Discharge State Selection, and Convergence Assessment

The scalar functions in Text S2 were evaluated over $0.01 \leq H \leq 1.99$. The semi-infinite integrals I_D and I_N were truncated at $k_{max} = 12$ and computed using adaptive quadrature with relative and absolute tolerances of 10^{-10} and 10^{-12} , respectively.

Both integrands possess removable limiting forms at $k = 0$. These limits were evaluated analytically as

$$\lim_{k \rightarrow 0} \frac{k^3}{\tanh(\alpha k) \sinh^2(\pi k)} = \frac{1}{\alpha \pi^2}, \quad \lim_{k \rightarrow 0} \frac{k \tanh(\alpha k)}{\sinh^2(\pi k)} = \frac{\alpha}{\pi^2} \quad (S27)$$

136 The limiting expressions in Eq. (S27) were used in the numerical implementation to avoid loss of
 137 accuracy near the lower integration boundary.

138 Algebraic consistency was assessed over 401 uniformly spaced filling ratios. The following residuals
 139 were evaluated:

$$\begin{aligned}
 140 \quad \varepsilon_1 &= Q_{FS} - (Q_D + Q_C + Q_R), \\
 141 \quad \varepsilon_2 &= S_B - (Q_{FS} - Q_D), \\
 142 \quad \varepsilon_3 &= \Pi_B - |S_B|(1 - \rho_B^2), \\
 143 \quad \varepsilon_4 &= \beta_B^2 + m_B^2 - 1. \tag{S28}
 \end{aligned}$$

144 All four residuals remained at the level of numerical roundoff over the investigated filling range.

145 An independent convergence assessment was conducted at 39 filling ratios over $0.05 \leq H \leq 1.95$. For
 146 this assessment, the spectral truncation limit was increased from $k_{max} = 12$ to $k_{max} = 16$, while the
 147 relative and absolute quadrature tolerances were reduced to 10^{-12} and 10^{-14} , respectively. The resulting
 148 values of Q_{RL} , Q_{FS} , Π_B , and Π_Q were compared with those obtained using the baseline numerical settings.
 149 The differences did not affect any value reported to the precision used in the main manuscript.

150 The half-filled state was evaluated independently using the stricter quadrature settings. The calculation
 151 recovered

$$152 \quad Q_D = Q_{FS} = \frac{\pi}{16}, \quad R_Q - C_Q = 0, \quad \beta_B = 0, \quad m_B = 1, \tag{S29}$$

153 to numerical roundoff, thereby reproducing the exact relations derived in Texts S1 and S2.

154 The characteristic filling states corresponding to the maxima of Q_{FS} , Π_B , and Π_Q were determined by
 155 bounded one-dimensional optimization of the exact scalar discharge and diagnostic functions using
 156 golden-section search and parabolic interpolation. The extrema were located independently of the plotting

grid. A parameter tolerance of 10^{-10} was used in the optimization, together with the stricter quadrature settings. The optimized filling ratios and diagnostic maxima remained unchanged to the precision reported in the main manuscript. These tests demonstrate that the stated ordering of the characteristic states is insensitive to the plotting resolution, spectral truncation, and quadrature tolerances used in the baseline calculations.

The equal-discharge comparison was constructed from the nonmonotonic physical-discharge relation. After locating the maximum-discharge filling ratio H_Q , the target discharge was defined as 95% of the maximum value:

$$Q_t = 0.95Q_{\text{FS}}(H_Q), \quad Q_{\text{FS}}(H) - Q_t = 0 \quad (\text{S30})$$

The two roots of Eq. (S30) were determined separately on the ascending interval $0.01 \leq H \leq H_Q$ and the descending interval $H_Q \leq H \leq 1.99$ using bracketed root finding with the stricter quadrature settings. The calculation yielded $H = 1.5663$ and $H = 1.8617$, with $Q_{\text{FS}} = 0.47502$ for both states. The relative discharge difference between the two solutions was less than 10^{-8} , and the discharge-closure residual $Q_{\text{FS}} - (Q_D - C_Q + R_Q)$ was less than 10^{-9} for each state.

The dimensionless wall shear was evaluated from the analytical wall-normal derivative of the exact physical velocity at the wetted circular wall. At $s = 1$, the resulting expression is

$$T_w(\xi) = -\frac{\partial U_{\text{FS}}}{\partial n} \Big|_{\Gamma_w} = \frac{1}{2} - (\cosh \xi + \cos \alpha) \cos \alpha \int_0^\infty \frac{k \cos(k\xi)}{\sinh(\pi k) \cosh(\alpha k)} dk. \quad (\text{S31})$$

The wall coordinate was converted to cumulative arclength and normalized by the total wetted perimeter:

$$\frac{\ell(\xi)}{P_w} = \frac{1}{P_w} \int_{-\infty}^{\xi} \left[\left(\frac{dX_w}{d\eta} \right)^2 + \left(\frac{dY_w}{d\eta} \right)^2 \right]^{1/2} d\eta, \quad P_w = \ell(\infty). \quad (\text{S32})$$

For the wall-shear calculations shown in Fig. 6(c), ξ was discretized using 501 points over $-9 \leq \xi \leq 9$, and the spectral integral was evaluated using 1,800 points over $10^{-5} \leq k \leq 14$. The calculations

178 reproduced left–right symmetry of the velocity and wall-shear fields to a relative tolerance of 10^{-9} . The
179 integrated wall shear satisfied the cross-sectional momentum balance to within a relative error of
180 5×10^{-3} .